SIGN RECOGNITION

A landmark-first model for Indian Sign Language fingerspelling

FROM CAMERA FRAME TO LETTER

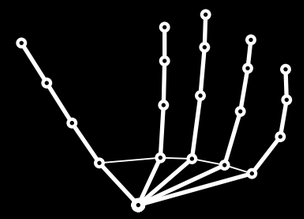
See the shape. Read the sign.

MediaPipe tracks hand joints;
a compact MLP classifies the pose.

21 POINTS / HAND

UP TO TWO HANDS

134 INPUT FEATURES



21-LANDMARK SKELETON

MODEL + HAND-COUNT GATE

134 → 256 → 128 → 64 → 26

21 joint points per hand; up to two hands.

1 HAND → 1-HAND TRAINING SUPPORT

2 HANDS → 2-HAND TRAINING SUPPORT

A class stays eligible when at least 25% of its training examples match the detected hand count.

17 epochs / best-loss checkpoint at epoch 11

26 static letters, including J and K

THE RECOGNITION LOOP

01 CAMERA → 02 MEDIAPIPE → 03 134 FEATURES → 04 MLP → 05 A-Z

HELD-OUT EVALUATION / 3,370 SOURCE IMAGES

76.38%	72.11%	55.07%
ACCURACY WHEN DETECTED	HAND COVERAGE	COVERAGE-ADJUSTED

2,430 of 3,370 held-out images had detected landmarks. Test data is not signer-separated.

ISL SIGN RECOGNITION / HIGH-CONTRAST PRINT EDITION

Static alphabet prototype - not connected signing or sentence translation.

If MediaPipe misses a hand, the count can be wrong. Treat output as a suggestion, not a guarantee.